

Lesson 2 · Turning Points and Optimisation

Learning intention

Use $f'(x) = 0$ to find turning points, decide maximum or minimum, find where a function increases or decreases, and solve maximum and minimum problems.

Success looks like

- I can find a turning point and justify its nature.
- I can give an interval where f is increasing or decreasing.
- I can set up and solve an optimisation problem, with units.

Slide 1 · Lesson 2 · Turning Points and Optimisation

Teaching line, not a question slide.

Timing: Do Now 10 min, three rounds about 20 min each, practice sheet 30 min, wrap-up 5 min.

Do Now

- 1 Differentiate $y = (x - 3)(x + 5)$.
- 2 Find the gradient of $f(x) = x^3 - 5x$ at the point where $x = 2$.
- 3 Solve $3x^2 - 12 = 0$.
- 4 Solve $2x^2 - x - 6 = 0$.

Slide 2 · Do Now

Q1 $\frac{dy}{dx} = 2x + 2$

Q2 7

Q3 $x = 2$ or $x = -2$

Q4 $x = 2$ or $x = -1.5$

Short solution:

Q1 expand: $y = x^2 + 2x - 15$

Q2 $f'(x) = 3x^2 - 5$; $f'(2) = 12 - 5$

Q3 $x^2 = 4$

Q4 $(2x + 3)(x - 2) = 0$

Watch for:

Q3 missing the negative root

Q4 sign of the -1.5 root

Diagnostic:

Q1 or Q2 wrong: Lesson 1 not secure; re-teach the power rule before round 1

Q3 wrong: expect missing turning points today (both roots of $f'(x) = 0$)

Q4 wrong: let the student use EQUA for $f'(x) = 0$ today

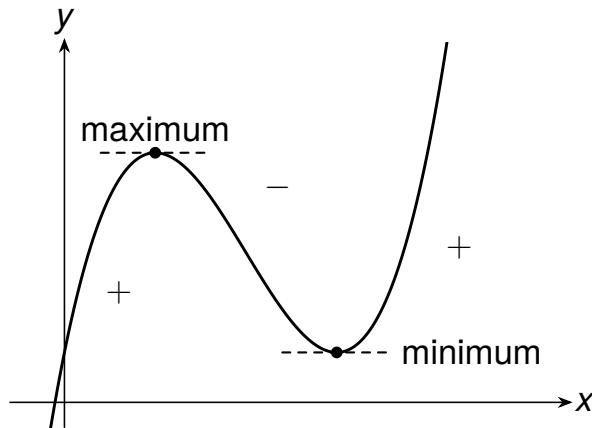
The one idea

At a turning point the tangent is flat:

$$f'(x) = 0$$

Maximum: gradient $+$ then $-$.

Minimum: gradient $-$ then $+$.



Slide 3 · The one idea

Teaching line, not a question slide.

Trace the curve with a finger: up, flat, down, flat, up.

Ask: what is the gradient exactly at the top? Zero, because the tangent is horizontal.

Finding turning points

Write it in this order

1. Differentiate.
 2. Set $f'(x) = 0$; solve for x .
 3. Find each y from $f(x)$.
 4. Maximum or minimum? Say why.
- 1** Find the turning points of $f(x) = x^3 - 6x^2 + 9x + 1$ and state their nature.

Slide 4 · Finding turning points

Q1 maximum (1, 5); minimum (3, 1)

Short solution:

Q1 $f'(x) = 3x^2 - 12x + 9 = 3(x - 1)(x - 3)$; $x = 1$ or $x = 3$

Q1 $f(1) = 5$, $f(3) = 1$

Q1 $f''(x) = 6x - 12$: $f''(1) = -6 < 0$ max; $f''(3) = 6 > 0$ min

Watch for:

stopping at $x = 1$, $x = 3$ (L1)

y -values from $f'(x)$ give 0, a sign of the wrong function

Maximum or minimum?

test	what you see	conclusion
gradient each side	$+\rightarrow 0\rightarrow -$	
	$-\rightarrow 0\rightarrow +$	
second derivative	$f'(x) < 0$	
	$f'(x) > 0$	

$f'(x)$: differentiate twice. Always **write the reason** in a sentence.

Slide 5 · Maximum or minimum?

Row 1 maximum

Row 2 minimum

Row 3 maximum, bends down

Row 4 minimum, bends up

Short solution:

sentence frame: $f''(1) = -6 < 0$, so the turning point at (1, 5) is a maximum

Watch for:

L2: a bare “maximum” with no test is not a justification

$f'' = 0$: the test fails; use the gradient either side instead

Practice

- 1 Find the turning point of $y = x^2 - 8x + 3$ and state whether it is a maximum or a minimum.
- 2 Find the turning points of $f(x) = 2x^3 - 3x^2 - 12x$ and justify their nature.
- 3 The curve $y = x^3 + kx$ has a turning point at $x = 2$. Find k .

Slide 6 · Practice

Q1 $(4, -13)$, minimum

Q2 maximum $(-1, 7)$; minimum $(2, -20)$

Q3 $k = -12$

Short solution:

Q1 $2x - 8 = 0$; $y = 16 - 32 + 3$; $y'' = 2 > 0$

Q2 $6x^2 - 6x - 12 = 6(x - 2)(x + 1)$; $f''(x) = 12x - 6$: $f''(-1) = -18$, $f''(2) = 18$

Q3 $3x^2 + k = 0$ at $x = 2$: $12 + k = 0$

Watch for:

Q2 forgetting to divide out the 6 is fine, but check both roots are found

Q3 is a Merit step: a turning point means $f' = 0$, not $f = 0$

Increasing and decreasing

increasing: $f'(x) > 0$ decreasing: $f'(x) < 0$

Find where $f'(x) = 0$ first; test the intervals between.

An up-parabola $f'(x)$ is **negative between its roots**.

2 For $f(x) = x^3 - 6x^2 + 9x + 1$, find where f is decreasing.

3 For which x is $y = 5 + 4x - x^2$ increasing?

Slide 7 · Increasing and decreasing

Q2 $1 < x < 3$

Q3 $x < 2$

Short solution:

Q2 $f'(x) = 3(x - 1)(x - 3)$, up-parabola, negative between 1 and 3

Q3 $\frac{dy}{dx} = 4 - 2x > 0$

Watch for:

L3: an answer like “ $x = 1$ and $x = 3$ ” is the turning points, not an interval

Q3 dividing by a negative flips the inequality if the student writes $-2x > -4$

Practice

- 1 For which values of x is $f(x) = x^2 - 10x + 1$ decreasing?
- 2 For which values of x is $f(x) = x^3 - 12x + 4$ decreasing?
- 3 Show that $f(x) = x^3 + 3x + 1$ is increasing for all values of x .

Slide 8 · Practice

Q1 $x < 5$

Q2 $-2 < x < 2$

Q3 $f'(x) = 3x^2 + 3 > 0$ for all x

Short solution:

Q1 $f'(x) = 2x - 10 < 0$

Q2 $f'(x) = 3x^2 - 12 = 3(x - 2)(x + 2) < 0$ between the roots

Q3 $x^2 \geq 0$, so $3x^2 + 3 \geq 3 > 0$

Watch for:

Q2 writing $x < 2$ only; the up-parabola is negative on a bounded interval

Q3 Merit: the reason ($x^2 \geq 0$) must be stated

Optimisation

Write it in this order

1. Formula in **one variable**.
2. Differentiate; set $= 0$; solve.
3. Justify maximum or minimum.
4. Answer the question asked, with units.

4 A ball's height is $h = 2 + 19.6t - 4.9t^2$ metres after t seconds. Find the maximum height.

Slide 9 · Optimisation

Q4 21.6 m

Short solution:

$$\text{Q4 } \frac{dh}{dt} = 19.6 - 9.8t = 0, t = 2$$

$$\text{Q4 } h = 2 + 39.2 - 19.6; \frac{d^2h}{dt^2} = -9.8 < 0$$

Watch for:

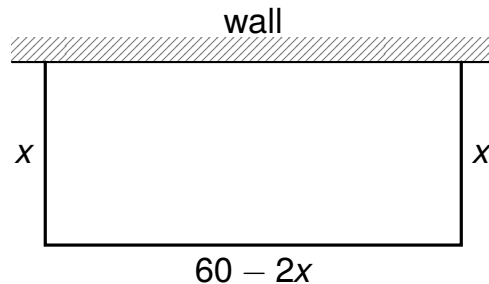
L5: answering $t = 2$ s instead of the height

L6: the unit (m)

Worked example · build the formula

5 A farmer has 60 m of fence to make a rectangular pen against a wall, fencing three sides. Find the maximum area.

Step 1: area in terms of x only.



Slide 10 · Worked example · build the formula

Q5 450 m^2 (pen 15 m by 30 m)

Short solution:

$$\text{Q5 } A = x(60 - 2x) = 60x - 2x^2$$

$$\text{Q5 } \frac{dA}{dx} = 60 - 4x = 0, x = 15; \frac{d^2A}{dx^2} = -4 < 0$$

Watch for:

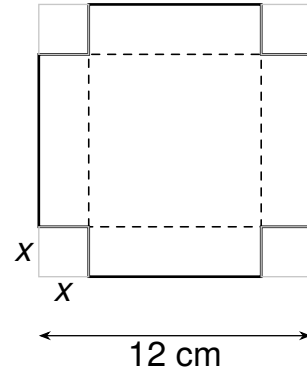
L4: writing $A = xy$ and differentiating with two letters

using $2x + 2y = 60$ (four sides) when the wall is one side

Practice

1 The profit, in dollars, from selling x items is $P = 120x - 2x^2 - 500$. How many items give the maximum profit, and what is that profit?

2 An open box is made from a 12 cm by 12 cm sheet by cutting a square of side x cm from each corner and folding up the sides. Find the value of x that gives the maximum volume, and the maximum volume.



Slide 11 · Practice

Q1 30 items; profit 1300 dollars

Q2 $x = 2$ cm; volume 128 cm^3

Short solution:

Q1 $P' = 120 - 4x = 0$; $P = 3600 - 1800 - 500$; $P'' = -4 < 0$

Q2 $V = x(12 - 2x)^2 = 4x^3 - 48x^2 + 144x$

Q2 $V' = 12x^2 - 96x + 144 = 12(x - 2)(x - 6)$; $x = 6$ gives no box

Q2 $V'' = 24x - 96 = -48 < 0$ at $x = 2$

Watch for:

Q2 Excellence: the model must be built and the impossible root rejected with a reason

Ways to lose marks

- L1 A turning point needs its y -value too.
- L2 “Maximum” needs a test and a sentence.
- L3 Increasing or decreasing is an interval.
- L4 One variable before you differentiate.
- L5 Answer the quantity asked, with units.

Slide 12 · Ways to lose marks

Teaching line, not a question slide.

Full list L1 to L6 in the handout; L6 (units and context) applies to every answer.

Summary

1. Turning point: $f'(x) = 0$; then y from $f(x)$.
2. Nature: $f'' < 0$ maximum, $f'' > 0$ minimum.
3. Increasing $f' > 0$; decreasing $f' < 0$: an interval.
4. Optimise: one variable, differentiate, $= 0$, justify, answer.

Slide 13 · Summary

Teaching line, not a question slide.

Ask the student to say each line back before looking at the screen.

Practice sheet · 30 minutes

Work alone on the practice sheet.

- Questions 1 and 2: warm-up, about 6 minutes.
- Questions 3 to 6: exam questions, about 20 minutes.
- Question 7: Extension, if time allows.

Every maximum needs a reason; every answer needs units.

Slide 14 · Practice sheet · 30 minutes

Q1 maximum $(-3, 64)$; minimum $(3, -44)$

Q2 12.5 m (at $t = 1.5$ s)

Q3 $-4 < x < 2.5$

Q4 26.8 m

Q5 (i) 352 and -352 viewers per month; (ii) 6336 viewers

Q6 wall height 3

Q7 (i) 169.8 cm^2 ; (ii) $A'' = -6\pi < 0$

Short solution:

Q3 $f'(x) = 2x^2 + 3x - 20 = (2x - 5)(x + 4)$, negative between the roots

Q4 $h' = 22.5 - 9.8t = 0$, $t = 2.296$; $h'' = -9.8 < 0$

Q5 (i) $t^2 - 48t + 320 = 0$, $t = 8$ or 40 ; $V' = -22t + 528$; (ii) $t = 24$

Q6 $A = 2x(9 - x^2) + x \cdot x^2 = 18x - x^3$; $x^2 = 6$; wall $9 - 6$

Q7 $L = 40 - \pi r$; $A = 2rL - \pi r^2 = 80r - 3\pi r^2$; $r = \frac{40}{3\pi}$

Watch for:

Q5 (i) two times give two rates, one positive, one negative

Q6 the triangle's height is x^2 , not 9

Full working: practice answer version (tutor's working, not an official mark scheme)

Homework

Homework sheet: 14 questions, about 60 minutes

- Questions 1 to 11: Achieved practice.
- Questions 12 and 13: Merit.
- Question 14: Extension (Excellence).

Optional extra: StudyTime Guide, “Calculating Turning Points” from page 16.

Slide 15 · Homework

Teaching line, not a question slide.

Answers: homework answer version (tutor’s working, not an official mark scheme).

Next lesson Do Now checks a turning point and a decreasing interval.